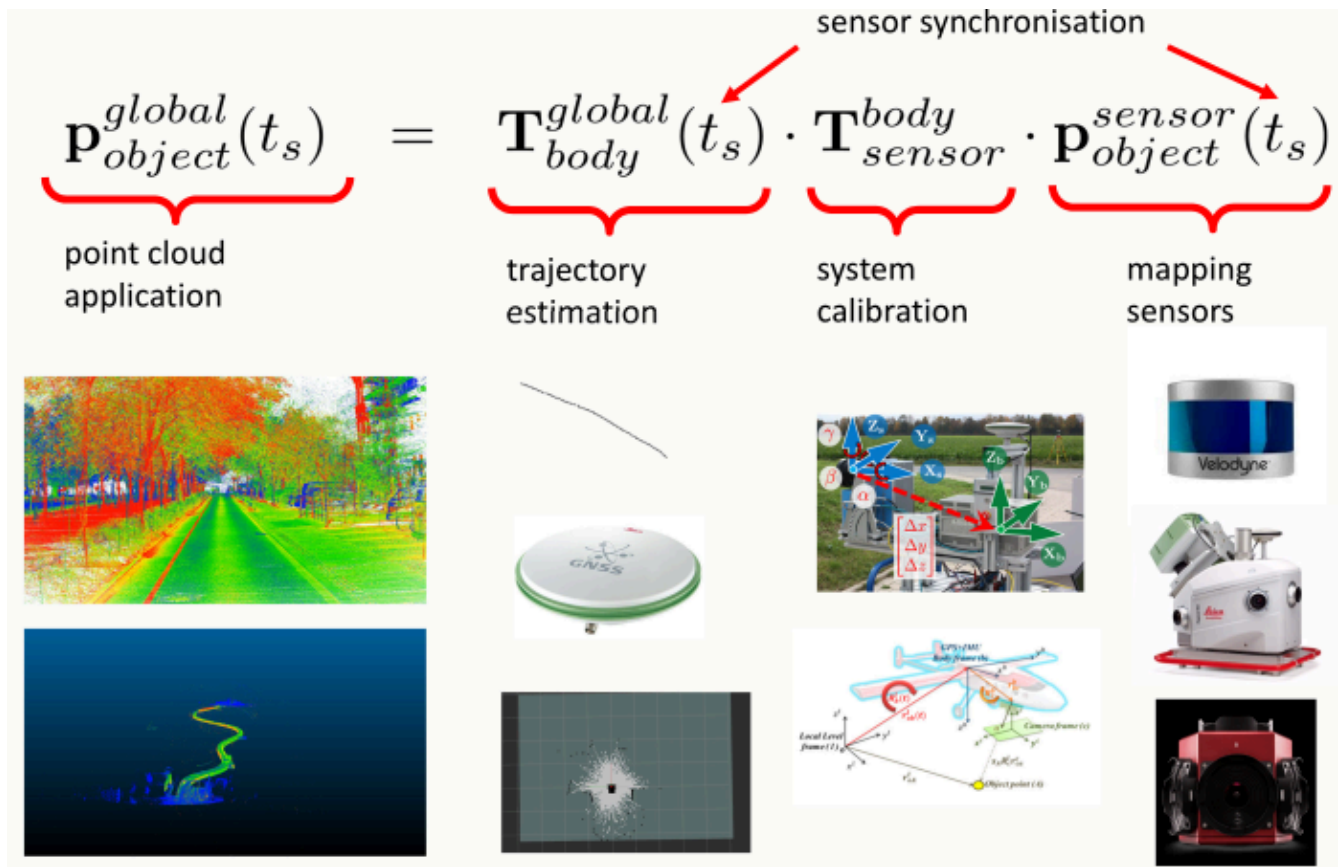


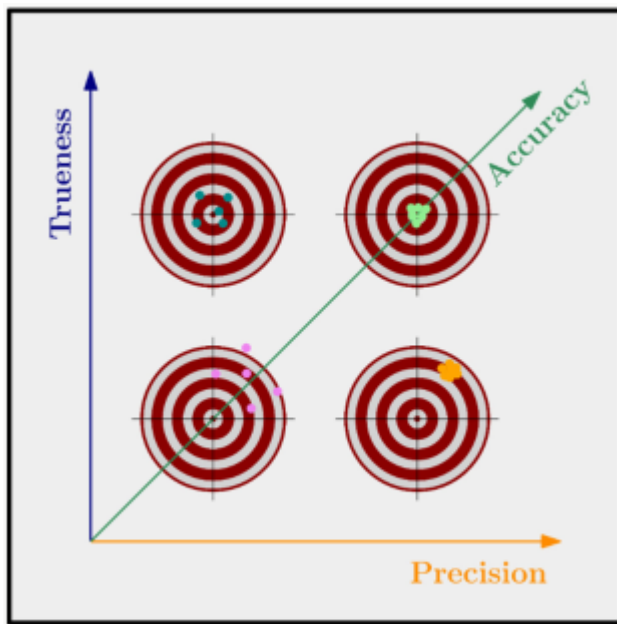
L8: Accuracy of Point Clouds



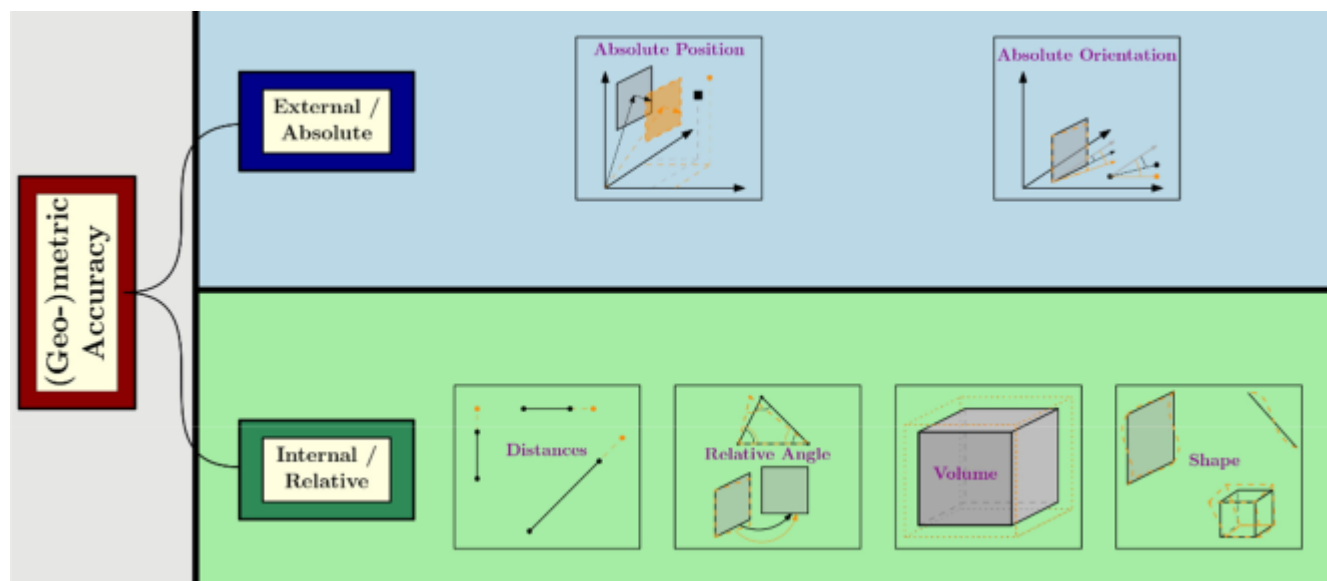
- Quality
- And Sufficiency for application

Accuracy

- **Accuracy:** Closeness of agreement between one test result the accepted reference value
- **Trueness:** Closeness of agreement between the **average** value obtained from a **large series** of test results and an accepted reference value
- **Precision:** Closeness of agreement between independent test results obtained under stipulated conditions



Geometric Accuracy



variances for each dimension

correlations

$$\Sigma_{pp} = \begin{bmatrix} \sigma_x^2 & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_y^2 & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_z^2 \end{bmatrix}$$

Covariance Matrix

Function relationship
between „inputs“ and „outputs“

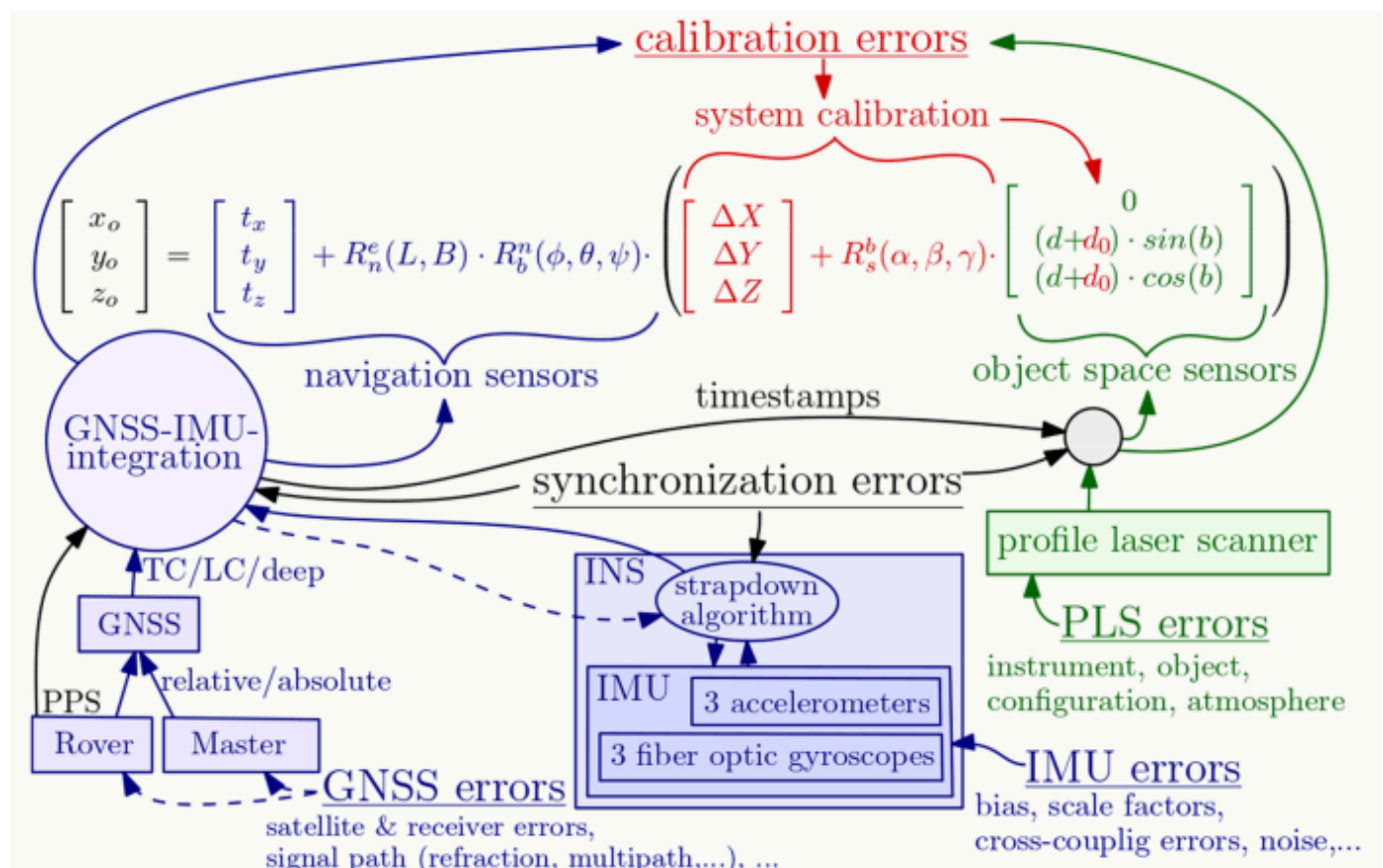
Covariance matrix
of inputs (observations)

$$\Sigma_{pp} = \mathbf{F} \Sigma_{ll} \mathbf{F}^T$$

Variance Propagation (Forward Modelling)

Simple Case of Covariance Propagation

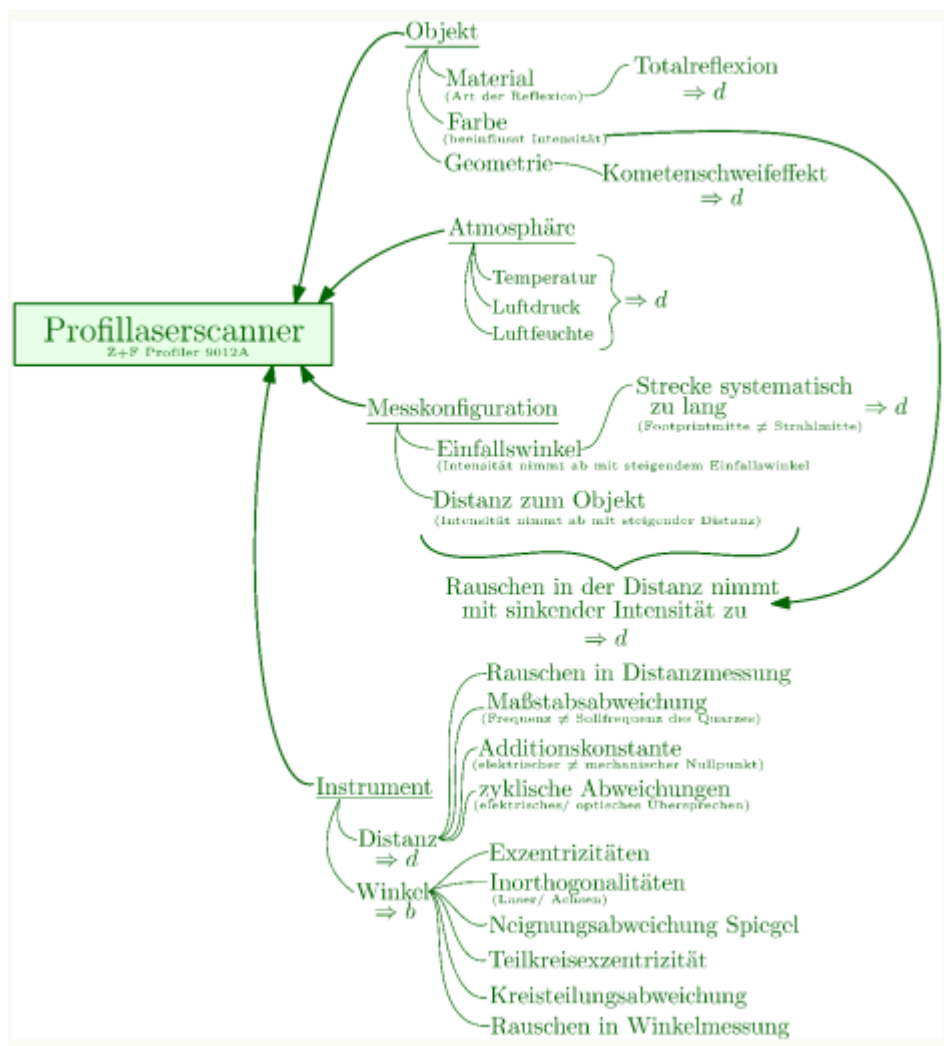
- Non-linear scalar function of 2 variable $f(a, b)$
- Uncertainty of input variables
 - $\Sigma_{ll} = \begin{bmatrix} \sigma_a^2 & \sigma_{ab} \\ \sigma_{ab} & \sigma_b^2 \end{bmatrix}$
- Uncertainty of result f
 - $\Sigma_{pp} = \sigma_f^2 = \left| \frac{\partial f}{\partial a} \right|^2 \sigma_a^2 + \left| \frac{\partial f}{\partial b} \right|^2 \sigma_b^2 + 2 \left| \frac{\partial f}{\partial a} \right| \left| \frac{\partial f}{\partial b} \right| \sigma_{ab}$
only if observations not independent
- Error propagation known from physics class



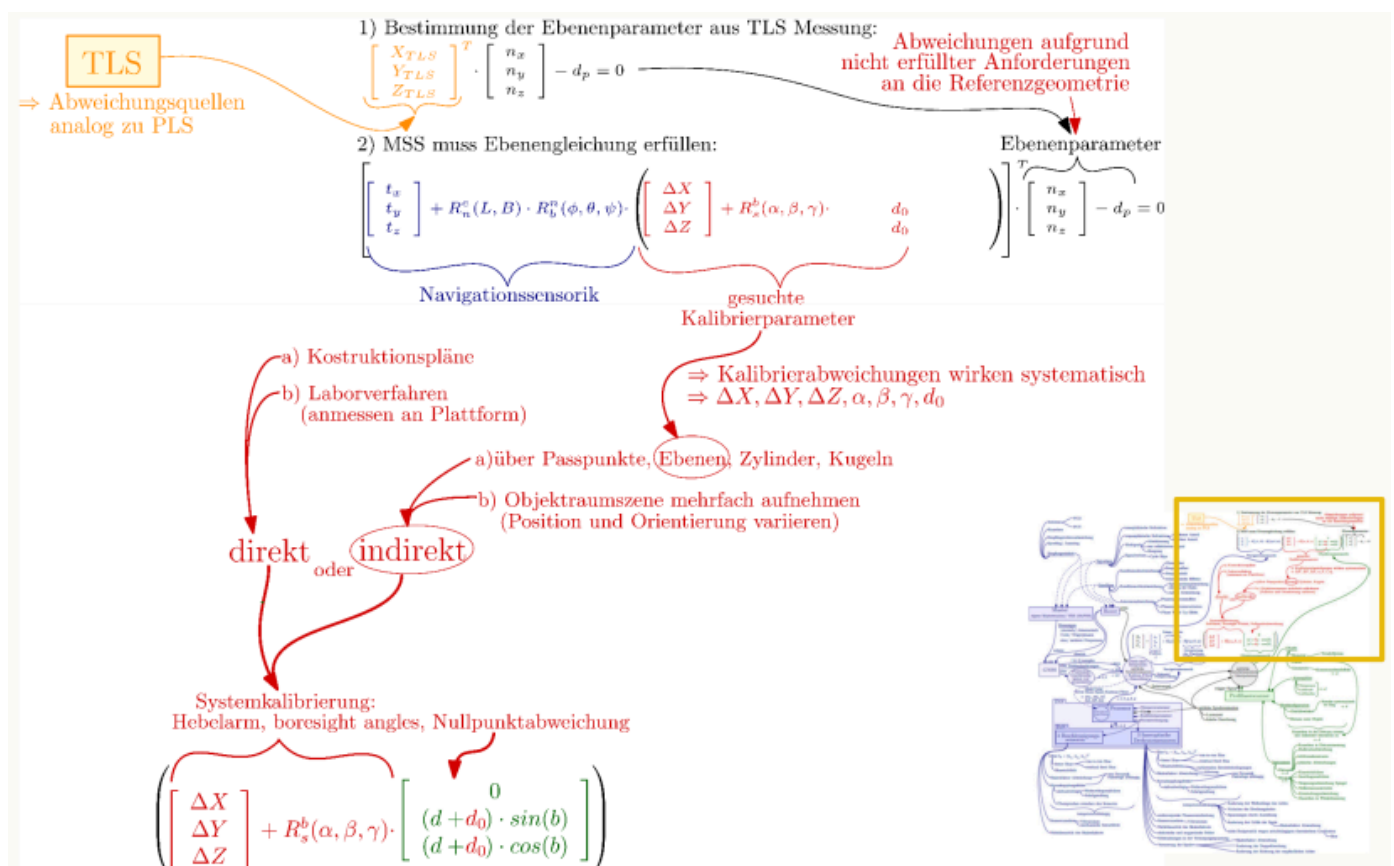
- **Functional relationships are often known**
- **But:**
 - They can be complex (GNSS-IMU-Integration)
 - Can be interdependent (system calibration)
 - Observations often already preprocessed (GNSS)
 - Sensor internal implementation often unknown (black box)

TLS Error

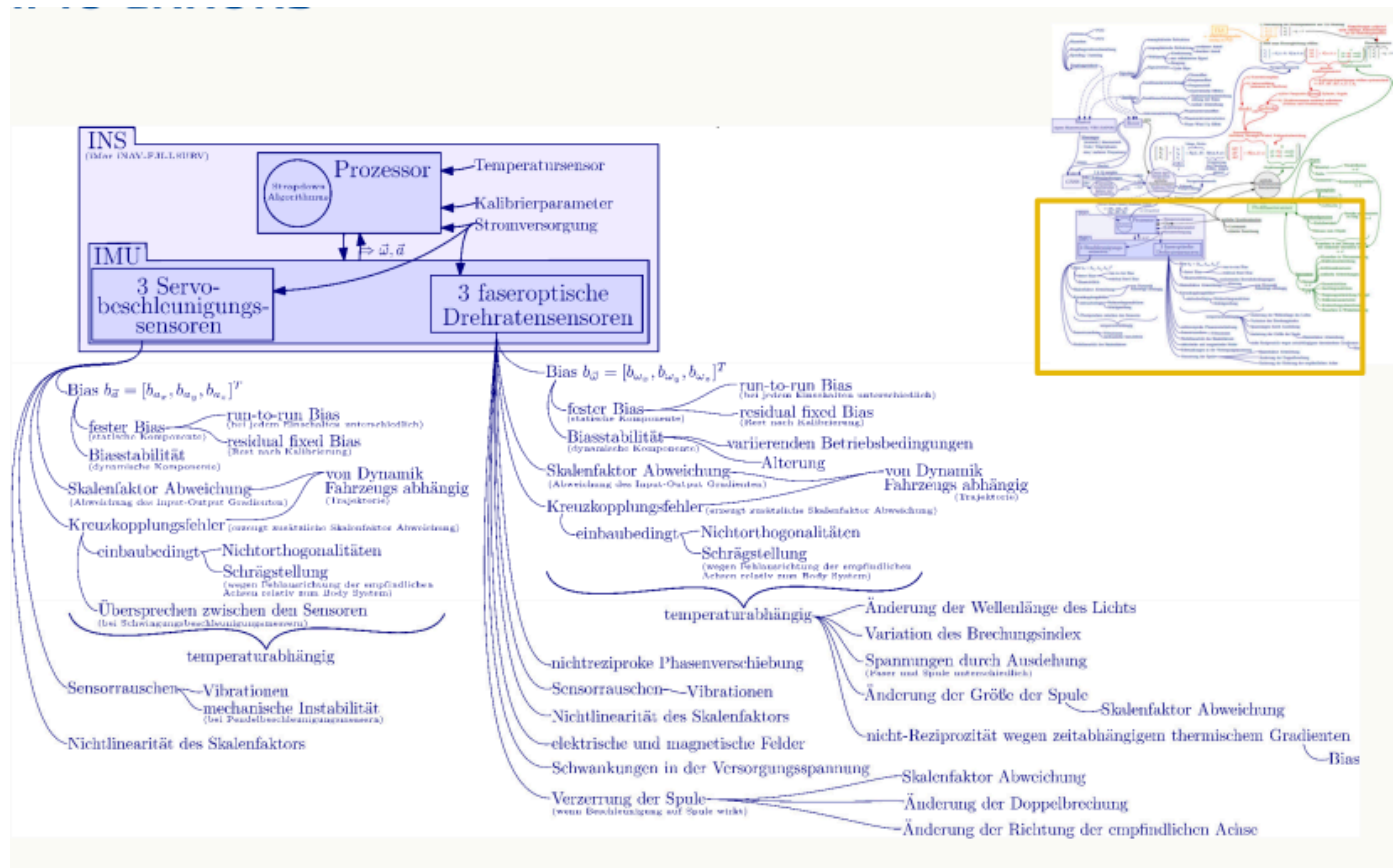
Category	Element	Random errors	Systematic errors	Outliers / Gross errors
Geometry	Distance	x		
	Incidence angle	x	x	
Object	Color	x	x	
	Material	x	x	Total reflection
	Form		x	Mixed Pixels



System Calibration Errors

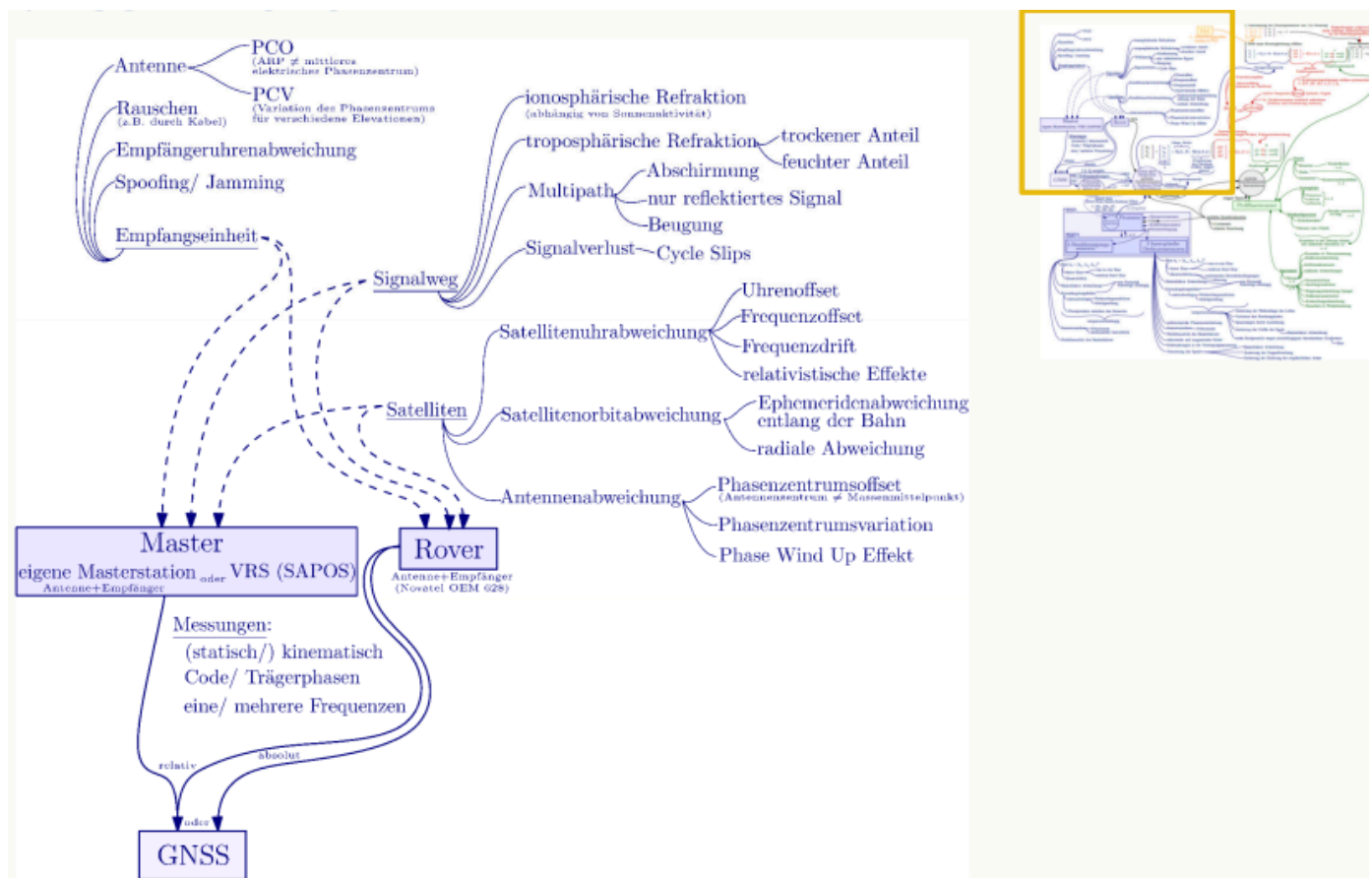


IMU Errors

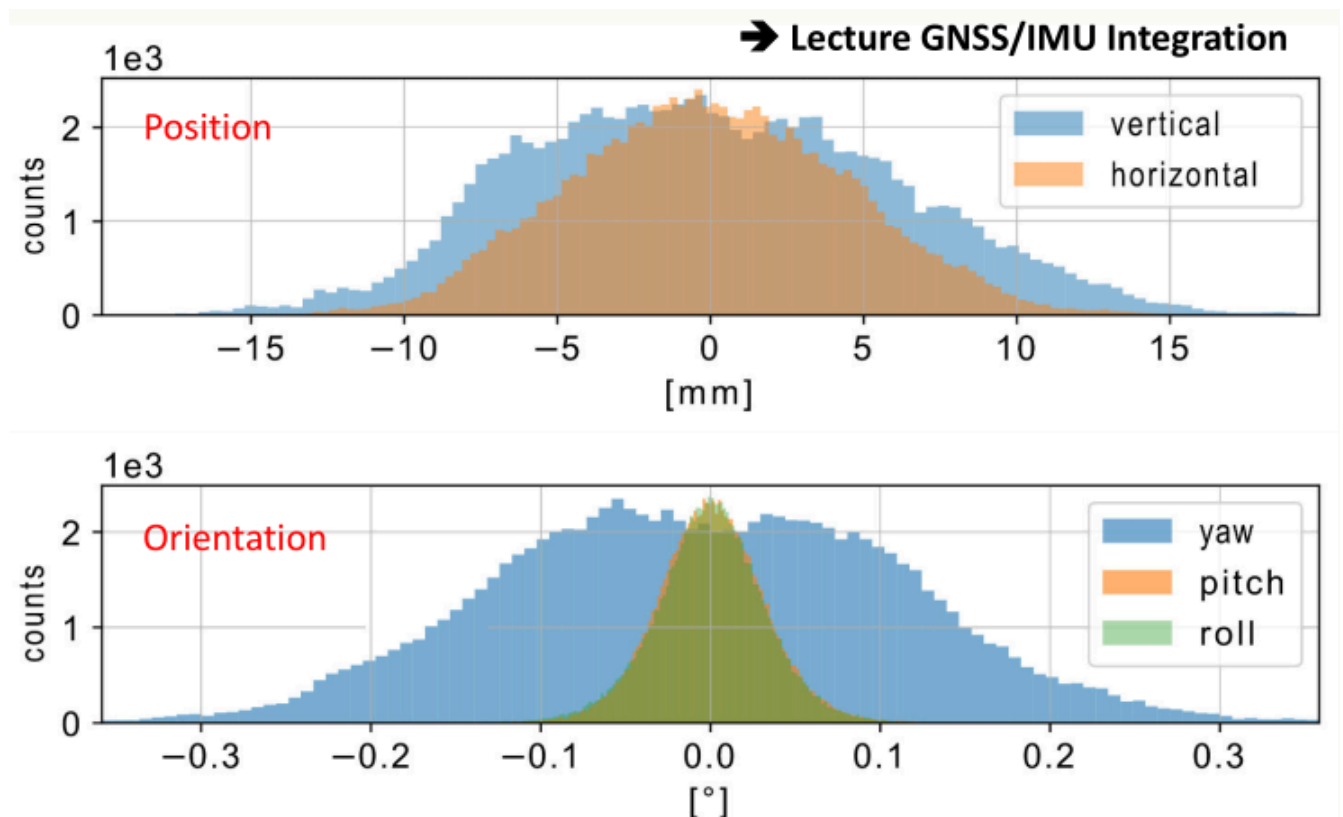


- INS Error
 - Constant Error
- INS Error Propagation
 - Noisy measurements of accelerometers and gyroscopes also lead to noisy position errors (uncertainty)

GNSS Errors



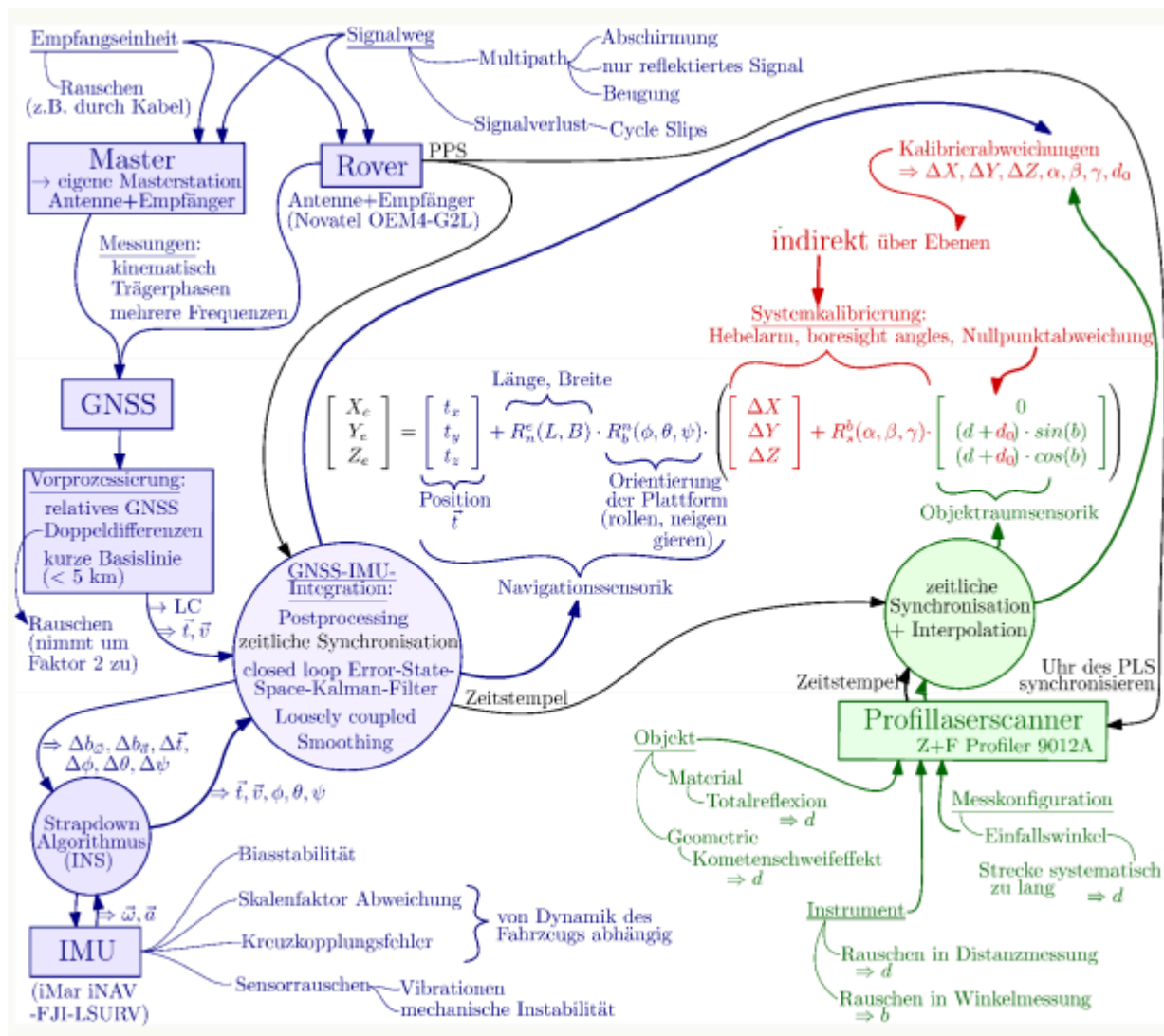
- GPS: Multipath: No white noise



Reduced Functional Model and Overview of Errors

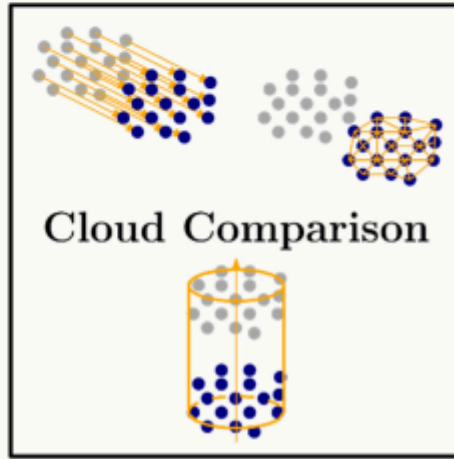
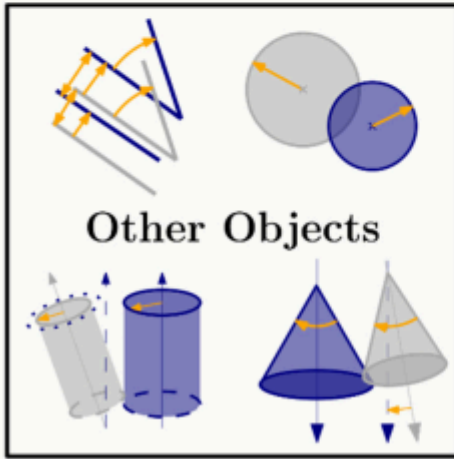
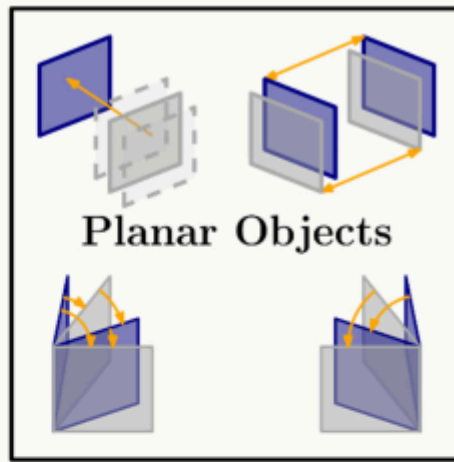
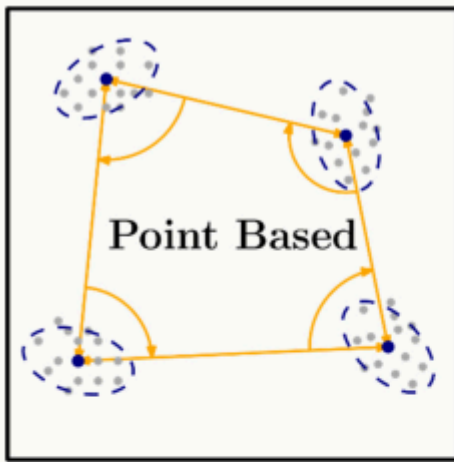
- Some errors can be eliminated by measurement/processing strategy (e.g. single-/double-differencing in GNSS)

- Some errors can be eliminated by calibration (e.g. IMU)
- Neglect black boxes and observation preprocessing



- **Alternative approach:** Empirical Determination on Σ_{pp}
- Multiple observations of a parameter
- Build the mean of parameter
- Calculate difference of individual parameter observations and the mean -> **precision**
- Compare the mean to a reference value -> **trueness/accuracy**
- Problem with laser scanning: You never observe the same point

Objects for Accuracy Analysis



Point Based Analysis

- Extract clearly identifiable points from the point cloud
- Compare target coordinates with reference values

Plane Based Analysis

- Extract facades and other planes
- Derive plane parameters
- Compare parameters from multiple passes and scanners
- Compare with reference

Other objects

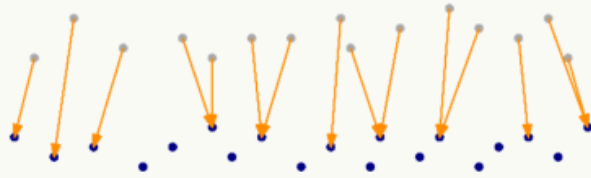
- lines
- cylinders

Cloud Comparisons

- So far:
 - Extract parameters from the point cloud
 - Point coordinates, plane parameters, radius of objects

- Compare parameters between multiple measurements and with reference

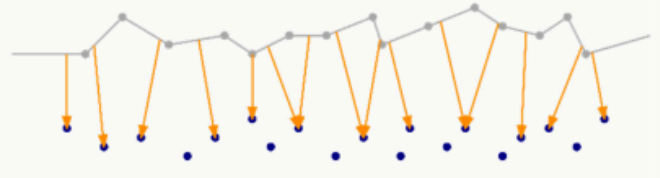
Cloud-2-Cloud (C2C)



Distance from Point in one Point Cloud to closest Point in other Point Cloud

➔ unsigned values

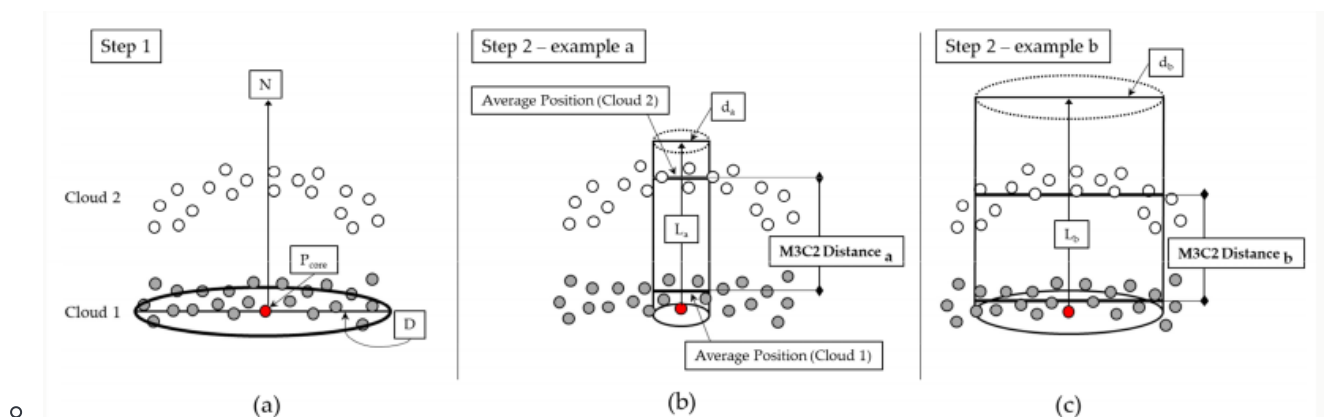
Cloud-2-Mesh (C2M)



Distance in Normal Direction from Mesh of one Point Cloud to closest Point in other Point Cloud

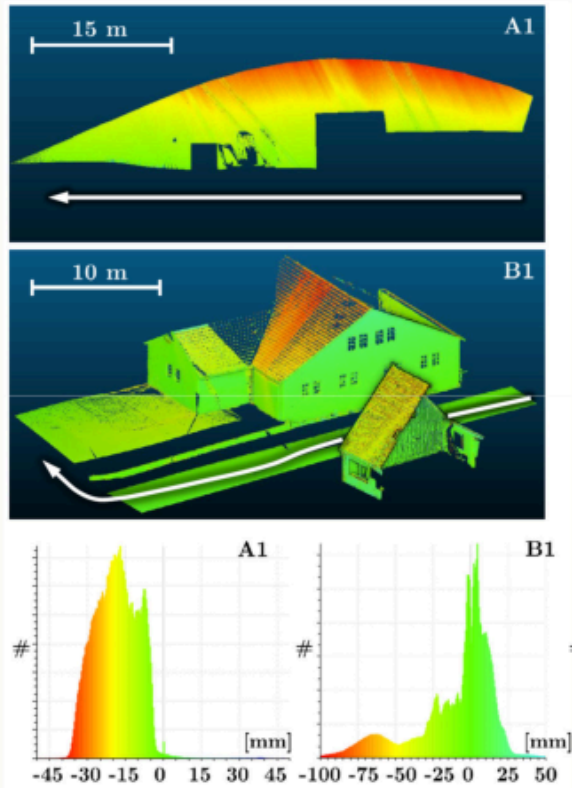
➔ signed values

- Multiscale Model-2-Model Cloud Comparison (M3C2)

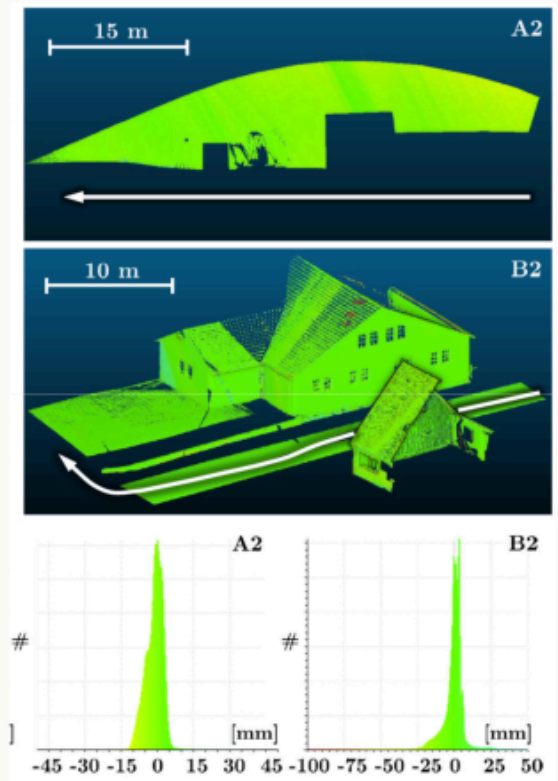


1. Select core point in cloud 1
2. Calculate normal vector at core point using neighbourhood radius $D/2$
3. Create cylinder ($r = d/2(\text{scale})$) along normal vector
4. Calculate average point for each cloud from points within the cylinder
5. M3C2 distance: distance between 2 average points

before



after



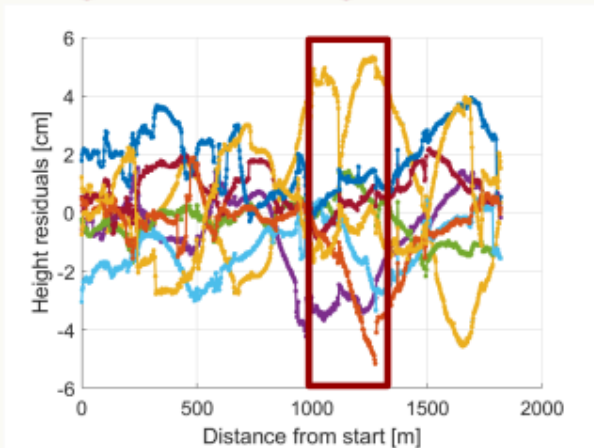
Reference point cloud from terrestrial laser scanner

- **Problems**

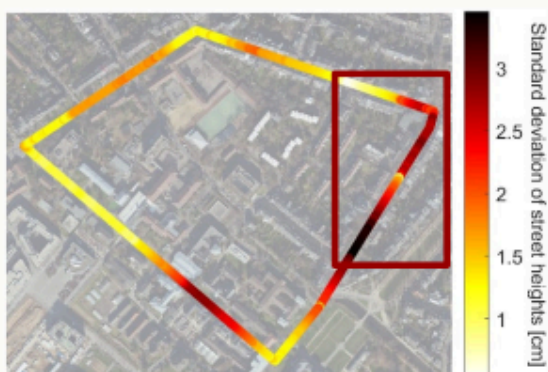
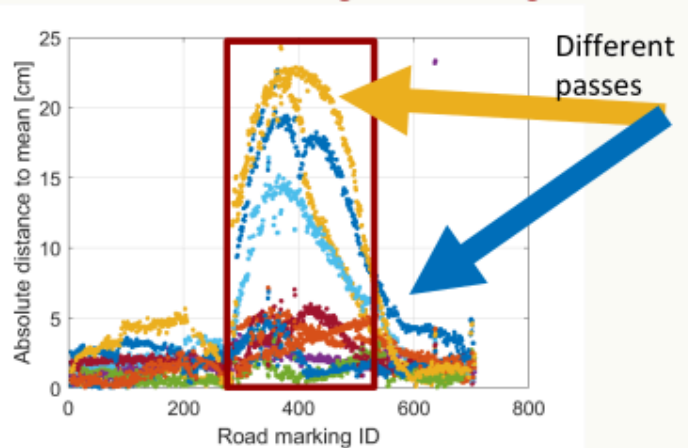
- Correspondences between points
- No differences visible for shifted planes
- Only curved structures show differences



Height evaluation using road surface



Position evaluation using road markings



Summary: Analyse Point Cloud Accuracy

1. Option: Forward modelling:

- Known:
 - All input value errors and distributions
 - All relationships between the observations and the output parameters (georeferenced 3D points)
- Forward model the influences of observation errors on to the parameter error

2. Option: **Empirical Analysis**

- Extract interesting parameters from the result point cloud
 - Feature point coordinates
 - Plane parameters
- Compare these parameters over multiple measurements and between reference parameters
- **Problems:**
 - Useful parameter selection
 - Reference Value derivation
 - Transferability of results:
 - locations & environments
 - Time periods
 - other applications
 - Determination of origin error from empirical parameter accuracies (**backward modelling**)